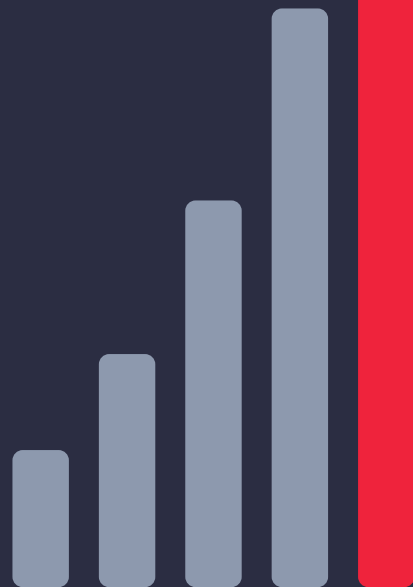


CO₂

Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Inequalities

Scope: 1760-2024

Data sources: Global Carbon Budget 2025 (via Our World in Data) • NASA Vital Signs



Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

Mauna Loa (NOAA), February 2026

Pre-industrial baseline: ~280 ppm

Today: >50% higher (~1.5×)

At least 1.5× the natural post-ice-age increase — but compressed into ~2 centuries.

Acceleration since 1958



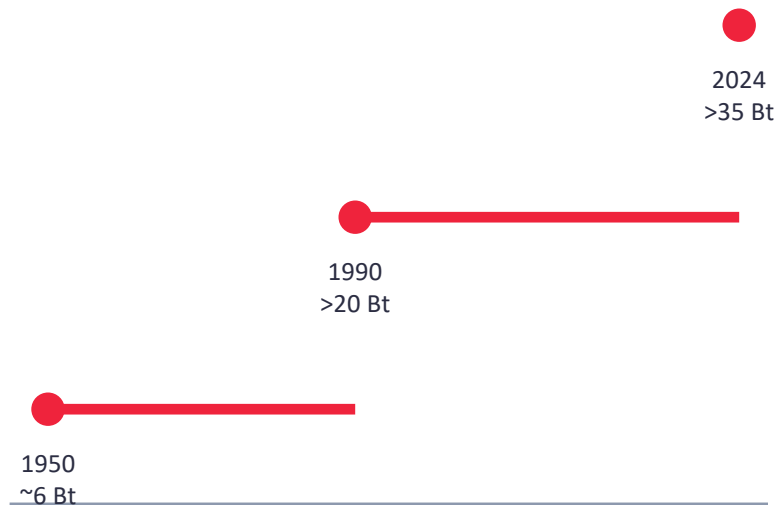
- Ground and satellite measurements confirm the sharp acceleration.
- Primary driver: burning coal, oil, and natural gas.

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

Inflection points

- 1950: ~6 billion tonnes (Bt)
- 1990: >20 Bt
- Today (2024): >35 Bt
- Growth slowed recently, but has not yet peaked

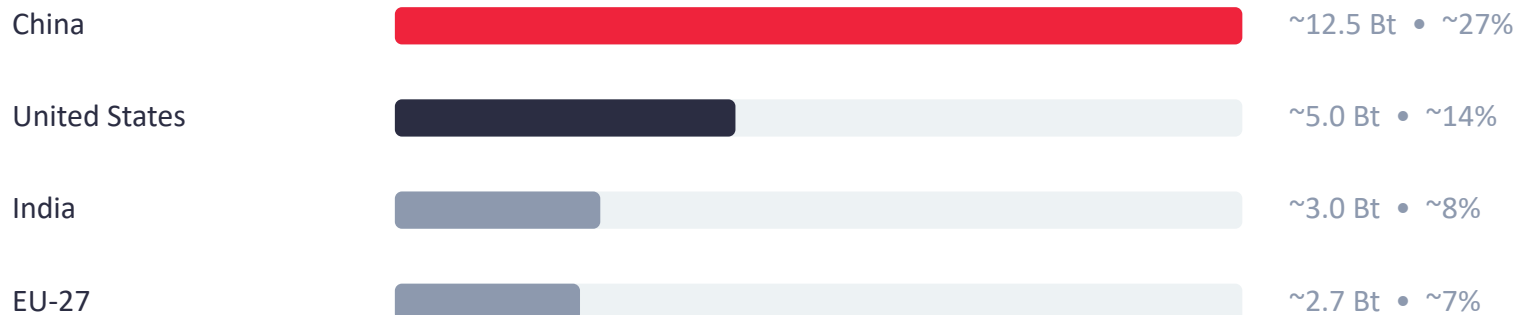
Trajectory (illustrative)



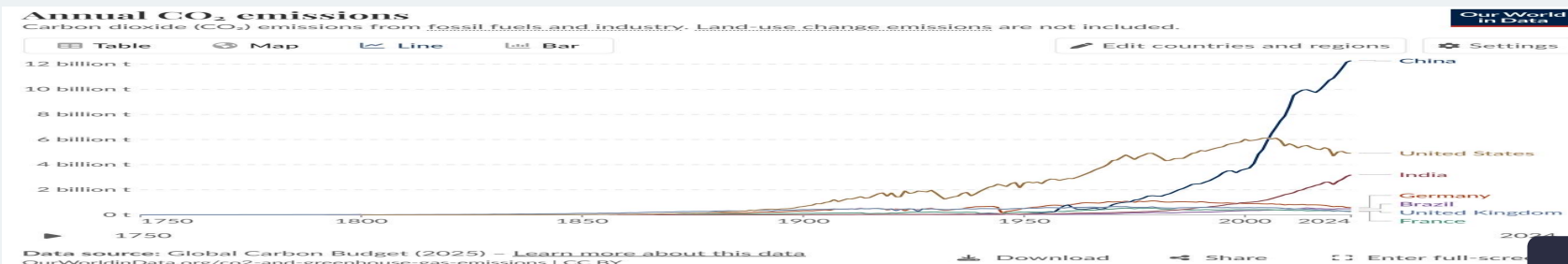
(Reference visual: figure_2.jpg)

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

Annual CO₂ emissions (fossil fuels + industry)



US peaked near ~6 Bt around 2005; now ~5 Bt (declining)



Source: Our World in Data — CO₂ and Greenhouse Gas Emissions (Global Carbon Budget 2025); see figure_3.jpg.
Reference visual: figure_3.jpg (Annual CO₂ emissions by country)

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

Regional share of annual CO₂ emissions (illustrative comparison)

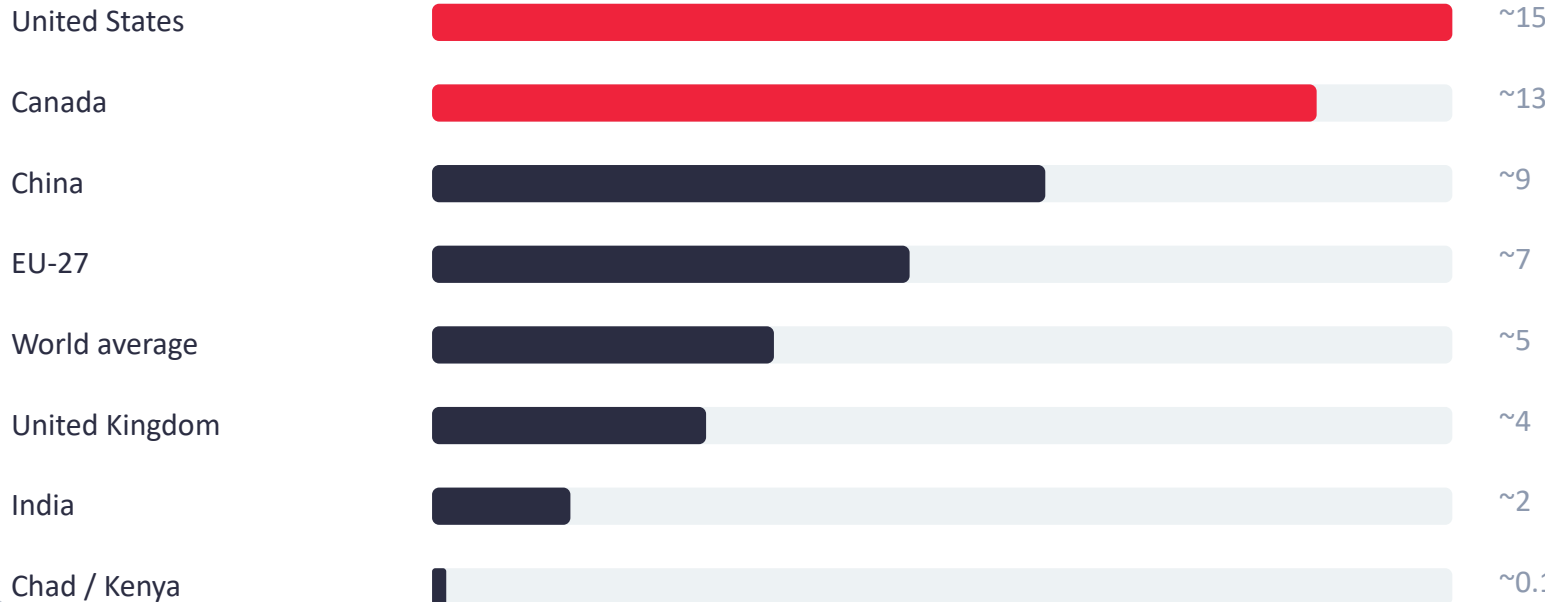


- 1900: Europe + the US accounted for >90% of global emissions
- 2024: Asia produces roughly half of emissions; US + Europe are below one-third.

Source: Our World in Data — CO₂ and Greenhouse Gas Emissions (Global Carbon Budget 2025).

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

Tonnes CO₂ per person per year (selected)



~150× gap: ~15 t/person (US/Australia) vs ~0.1 t/person (Chad/Niger)

Source: Our World in Data — CO₂ and Greenhouse Gas Emissions (Global Carbon Budget 2025); figure_1.jpg.

Ref:

6/9

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

Our World
in Data



Table



Map



Line



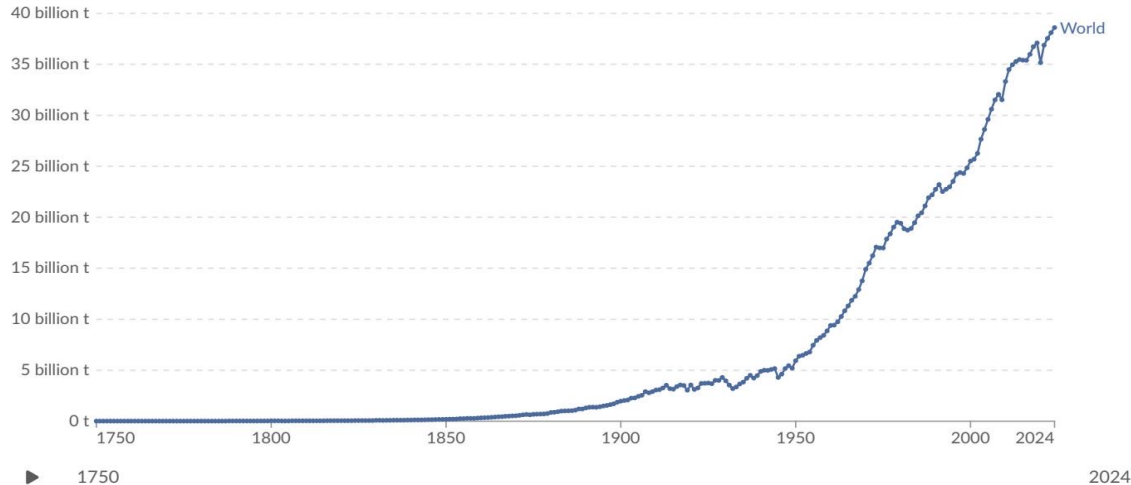
Bar



Edit countries and regions



Settings



Data source: Global Carbon Budget (2025) – [Learn more about this data](#)
OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY



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What the map shows

- US and much of Europe: declines (~-1% to -5%), shown in blue.
- Russia and parts of Southeast Asia: increases (+2% to +10% or more), shown in orange/red.
- Several African nations: increases (+2% to +10% or more).
- Some South American countries: very large percentage increases (>10%), while others declined.

Blue = negative change • Orange/Red = positive

Reference visual: [figure_4.jpg](#)

Source: Our World in Data — CO₂ and Greenhouse Gas Emissions (Global Carbon Budget 2025); [figure_4.jpg](#).

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

- Similar living standards can yield very different per-capita CO₂ depending on energy sources.
- France and the UK: higher nuclear and renewable shares → lower per-capita emissions.
- Germany and the Netherlands: higher fossil fuel share → higher per-capita emissions.

Comparison	Energy mix (high level)	Per-capita CO ₂ (direction)
France / UK	Higher nuclear + renewables	Lower

Policy implication: decarbonizing electricity supply can reduce per-capita CO₂ without changing prosperity.

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

>35 Bt and not yet peaked

Global fossil-fuel + industry CO₂ emissions exceed 35 billion tonnes per year; growth slowed but has not peaked.

China >25% of total

China emits ~12.5 Bt (~27%) — more than double the United States (~5 Bt). India (~3 Bt) is rising fast.

Regional center shifted

From >90% (Europe + US) in 1900 to ~50% in Asia today; US + Europe combined are below one-third.

Per-capita inequality (150×)

~15 t/person in the US/Australia vs ~0.1 t/person in Chad/Niger — inequality remains extreme.

Energy mix shapes outcomes

Policy and power-sector choices (nuclear/renewables vs fossil fuels) drive major per-capita differences.